ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

THE POISSON DISTRIBUTION

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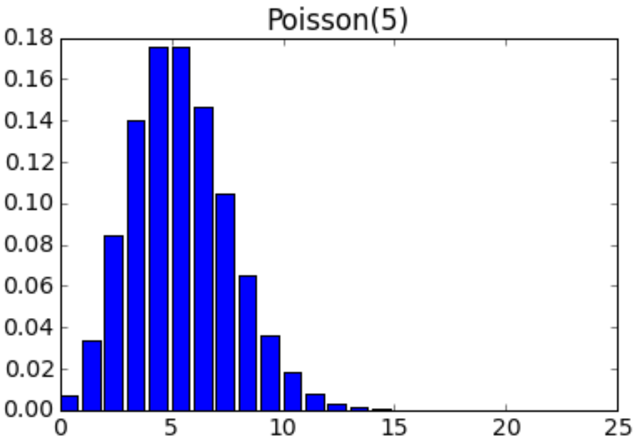
The Poisson Distribution

A distribution over the non-negative integers $\{0, 1, 2, \dots\}$



Poisson(λ), with $\lambda > 0$:

$$Pr(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}$$

- Mean: $\mathbb{E}X = \lambda$
- Variance: $\mathbb{E}(X - \lambda)^2 = \lambda$

How the Poisson Arises

Count the number of events (collisions, phone calls, etc) that occur in a certain interval of time. Call this number X, and say it has expected value λ.

Now suppose we divide the interval into small pieces of equal length.

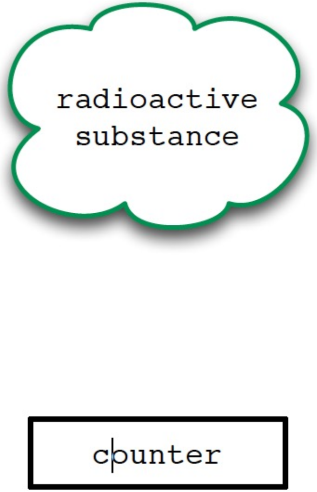
If the probability of an event occurring in a small interval is:

- independent of what happens in other small intervals, and
- the same across small intervals,

then $X \sim \text{Poisson}(\lambda)$.

Poisson: Examples

Rutherford's experiments with radioactive disintegration (1920)



- $N = 2608$ intervals of 7.5 seconds
- $N_k = \#$ intervals with k particles
- Mean: 3.87 particles per interval

k	0	1	2	3	4	5	6	7	8	≥ 9
N_k	57	203	383	525	532	408	273	139	45	43
P(3.87)	54.4	211	407	526	508	394	254	140	67.9	46.3

Flying bomb hits on London in WWII

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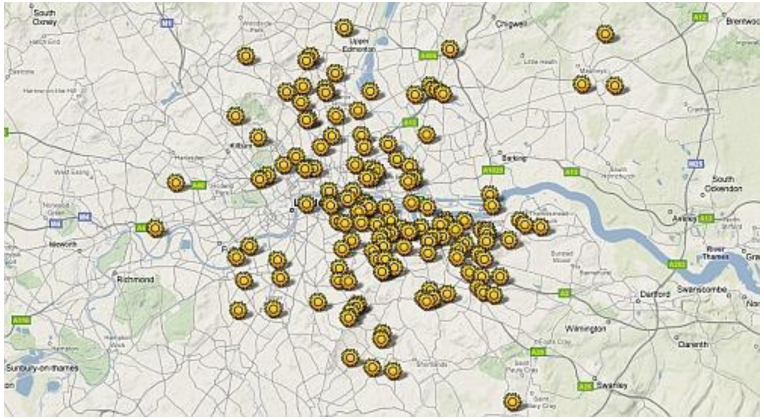

- Area divided into 576 regions, each $0.25\ \mathrm{km}^2$
- $N_k = \#$ intervals with k particles
- Mean: 0.93 hits per region

k	0	1	2	3	4	≥ 5
N_k	229	211	93	35	7	1
P(0.93)	226.8	211.4	98.54	30.62	7.14	1.57

Fitting a Poisson Distribution to Data

Given samples $x_1, \ldots, x_n$, what Poisson(λ) model to choose?